

Proportion

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Coordinate Geometry (ref `cg4`)

This document contains all 50 *Proportion* questions, each with a fully worked solution and a TikZ graph of the relationship: a straight line through the origin for direct proportion, a parabola for $y \propto x^2$, a cube curve for $y \propto x^3$, a square-root curve for $y \propto \sqrt{x}$, and a hyperbola for inverse proportion. The given and required data points are marked on each curve. (The two pure algebraic-proof questions have no graph.)

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Proportion 01 (cg4-001)

Difficulty: Foundation / Marks: 3

Given that $y \propto x$ and that $y = 45$ when $x = 9$, find y when $x = 5$.

Worked solution.

Step 1: Write the proportionality statement as an equation.

$$y = kx$$

$y \propto x$ means $y = kx$ for some constant k .

Step 2: Substitute the known values to find k .

$$45 = k \times 9 \implies k = 5$$

Divide both sides by 9.

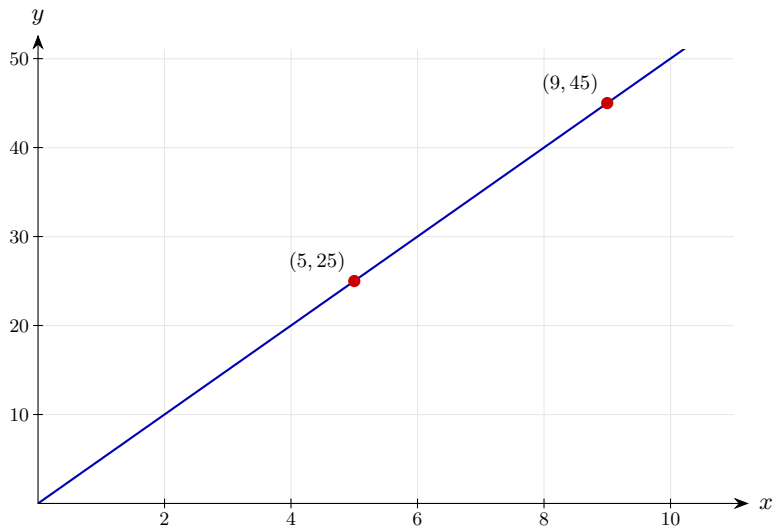
Step 3: Write the equation and substitute $x = 5$.

$$y = 5x \implies y = 5 \times 5 = 25$$

Use the complete equation to find the required value.

Final answer: $y = 25$

Graph.



The relationship plotted, with the data points marked.

Proportion 02 (cg4-002)

Difficulty: Foundation / Marks: 3

Given that $y \propto x$ and that $y = 36$ when $x = 4$, find the value of x when $y = 63$.

Worked solution.

Step 1: Write $y = kx$ and substitute the known pair.

$$36 = k \times 4 \implies k = 9$$

The constant of proportionality is 9.

Step 2: Form the equation.

$$y = 9x$$

Now we can use any value of x or y .

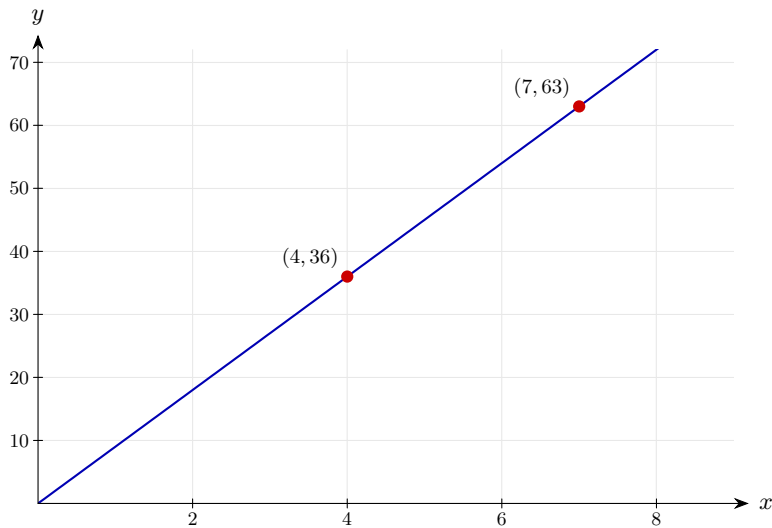
Step 3: Substitute $y = 63$ and solve for x .

$$63 = 9x \implies x = 7$$

Divide both sides by 9.

Final answer: $x = 7$

Graph.



The relationship plotted, with the data points marked.

Proportion 03 (cg4-003)

Difficulty: Foundation / Marks: 3

A car uses petrol directly proportional to the distance it travels. It uses 6 litres to travel 48 km. How many litres does it use to travel 200 km?

Worked solution.

Step 1: Let P litres of petrol and d km distance. Write $P = kd$.

$$6 = k \times 48 \implies k = \frac{6}{48} = \frac{1}{8}$$

Find the constant of proportionality.

Step 2: Write the equation.

$$P = \frac{1}{8}d$$

The car uses 1 litre per 8 km.

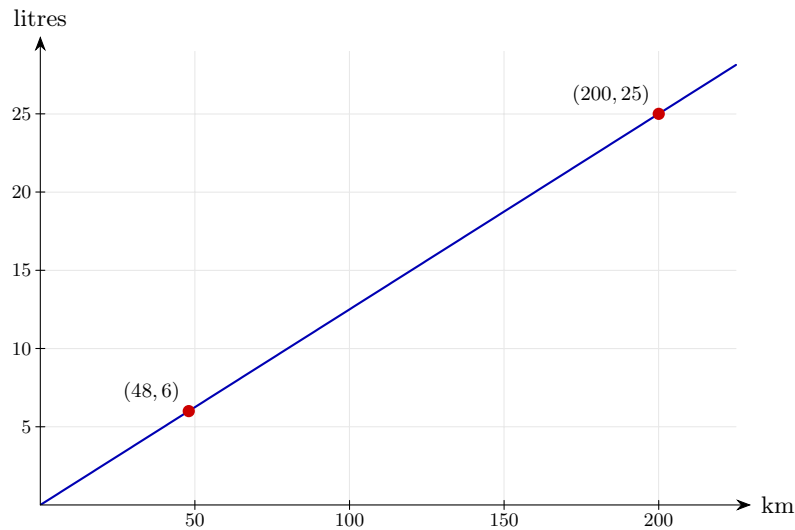
Step 3: Substitute $d = 200$.

$$P = \frac{200}{8} = 25 \text{ litres}$$

Multiply the distance by the rate.

Final answer: 25 litres

Graph.



The relationship plotted, with the data points marked.

Proportion 04 (cg4-004)

Difficulty: Foundation / Marks: 4

Given that $y \propto x$, that $y = 15$ when $x = a$, and that $y = a$ when $x = 60$, find the positive value of a .

Worked solution.

Step 1: Write $y = kx$ and use the first pair.

$$15 = ka \implies k = \frac{15}{a}$$

Express k in terms of a .

Step 2: Substitute the second pair $y = a$, $x = 60$.

$$a = k \times 60 = \frac{15}{a} \times 60 = \frac{900}{a}$$

Replace k with the expression found above.

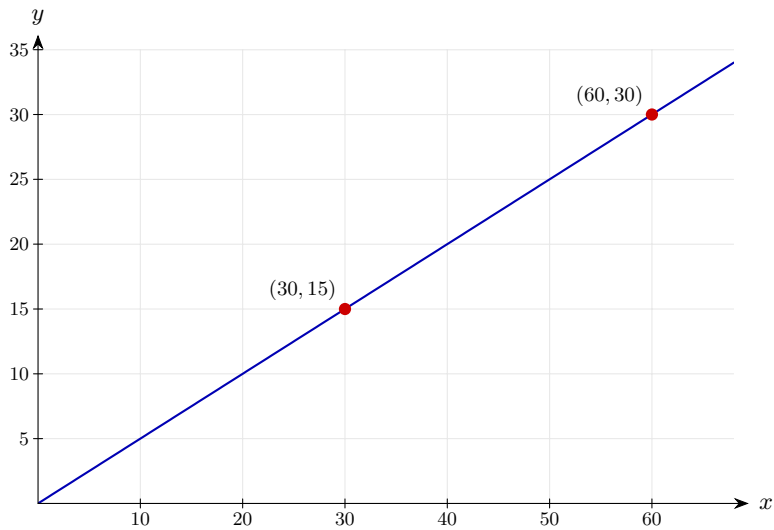
Step 3: Solve $a^2 = 900$.

$$a^2 = 900 \implies a = 30 \quad (a > 0)$$

Take the positive square root.

Final answer: $a = 30$

Graph.



The relationship plotted, with the data points marked.

Proportion 05 (cg4-005)

Difficulty: Foundation / Marks: 3

A plumber charges a call-out fee of £0 plus an amount directly proportional to the number of hours worked. She charges £84 for 3 hours. How much does she charge for 7 hours?

Worked solution.

Step 1: Let C = charge (£) and h = hours. Write $C = kh$.

$$84 = k \times 3 \implies k = 28$$

She charges £28 per hour.

Step 2: Form the equation.

$$C = 28h$$

The constant of proportionality is 28.

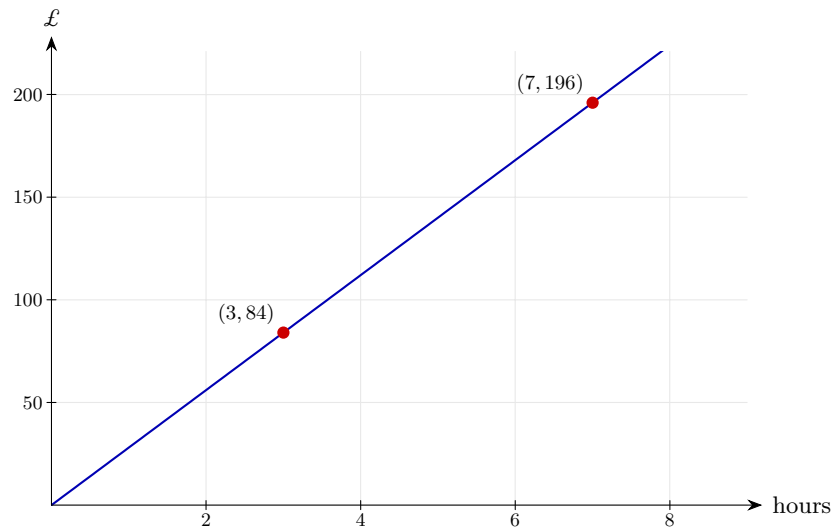
Step 3: Substitute $h = 7$.

$$C = 28 \times 7 = £196$$

Multiply hours by the rate.

Final answer: £196

Graph.



The relationship plotted, with the data points marked.

Proportion 06 (cg4-006)

Difficulty: Foundation / Marks: 3

Given that $y \propto x$, $y = 30$ when $x = 12$, and $y = 75$ when $x = a$, find a .

Worked solution.

Step 1: Find k using the first known pair.

$$k = \frac{30}{12} = \frac{5}{2}$$

The constant of proportionality is $\frac{5}{2}$.

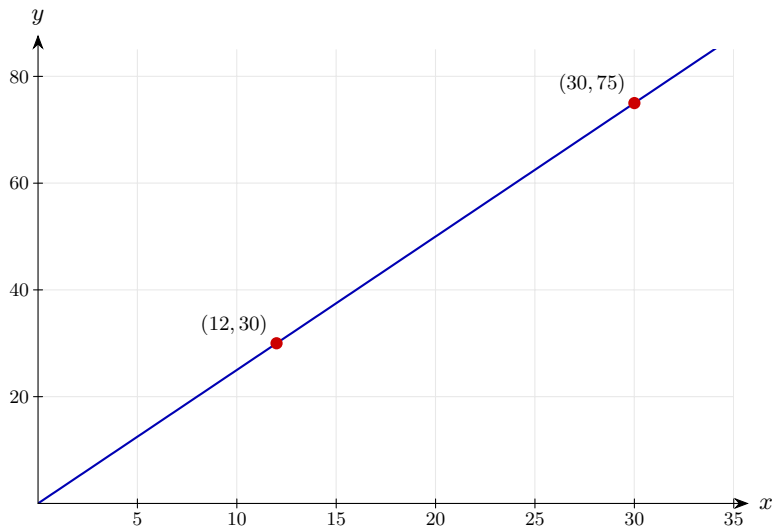
Step 2: Use $y = 75$ to find a .

$$75 = \frac{5}{2} \times a \implies a = \frac{75 \times 2}{5} = 30$$

Multiply both sides by $\frac{2}{5}$.

Final answer: $a = 30$

Graph.



The relationship plotted, with the data points marked.

Proportion 07 (cg4-007)

Difficulty: Foundation / Marks: 3

The cost C of printing leaflets is directly proportional to the number printed, n . It costs £12 to print 80 leaflets. Find the cost of printing 350 leaflets.

Worked solution.

Step 1: Write $C = kn$ and find k .

$$12 = k \times 80 \implies k = \frac{12}{80} = 0.15$$

Each leaflet costs £0.15 to print.

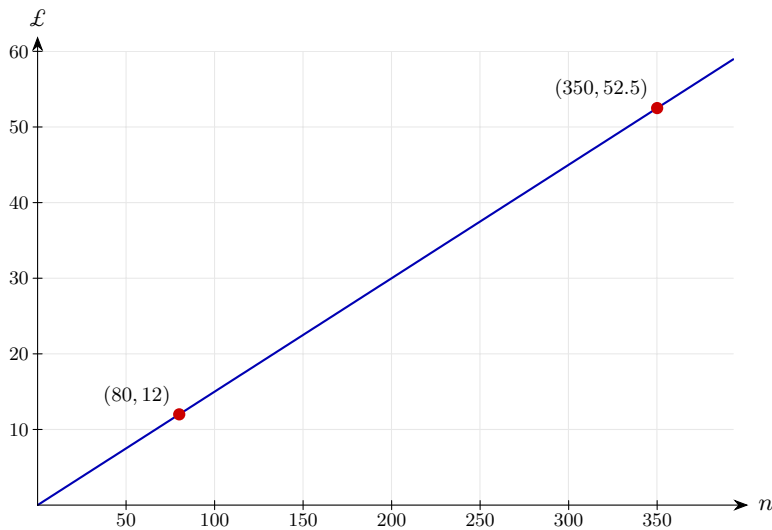
Step 2: Substitute $n = 350$.

$$C = 0.15 \times 350 = £52.50$$

Multiply number of leaflets by cost per leaflet.

Final answer: £52.50

Graph.



The relationship plotted, with the data points marked.

Proportion 08 (cg4-008)

Difficulty: Foundation / Marks: 5

The extension e cm of a spring is directly proportional to the force F N applied to it. The spring extends 6 cm when a force of 15 N is applied.

- Find the equation connecting e and F .
- Find the extension when a force of 40 N is applied.
- Find the force needed to extend the spring by 9 cm.

Worked solution.

Step 1: Write $e = kF$ and substitute the known values.

$$6 = k \times 15 \implies k = \frac{6}{15} = \frac{2}{5}$$

Find the constant of proportionality.

Step 2: State the equation (part a).

$$e = \frac{2}{5}F$$

This is the complete relationship between e and F .

Step 3: Part b: substitute $F = 40$.

$$e = \frac{2}{5} \times 40 = 16 \text{ cm}$$

The spring extends 16 cm.

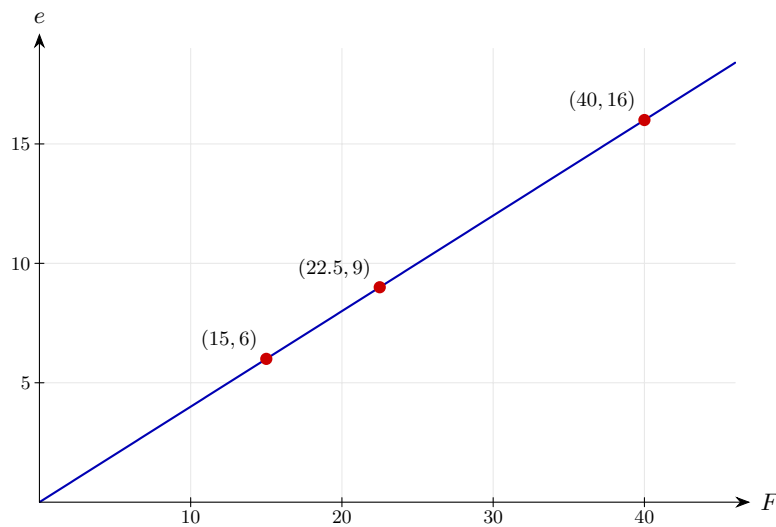
Step 4: Part c: substitute $e = 9$ and solve for F .

$$9 = \frac{2}{5}F \implies F = \frac{9 \times 5}{2} = 22.5 \text{ N}$$

Multiply both sides by $\frac{5}{2}$.

Final answer: a) $e = \frac{2}{5}F$; b) 16 cm; c) 22.5 N

Graph.



The relationship plotted, with the data points marked.

Proportion 09 (cg4-009)

Difficulty: Foundation / Marks: 3

Given that y is inversely proportional to x , and that $y = 8$ when $x = 5$, find y when $x = 20$.

Worked solution.

Step 1: Write the inverse proportion equation.

$$y = \frac{k}{x}$$

$y \propto \frac{1}{x}$ is equivalent to $y = \frac{k}{x}$.

Step 2: Find k using the known pair.

$$8 = \frac{k}{5} \implies k = 40$$

Multiply both sides by 5.

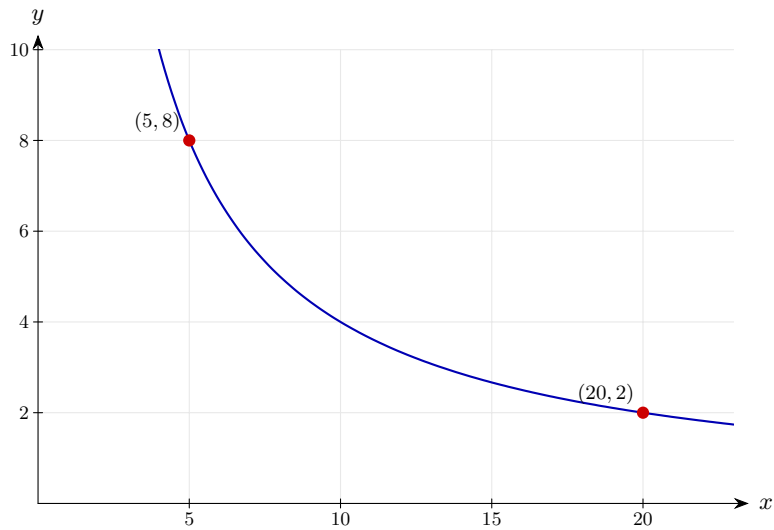
Step 3: Substitute $x = 20$.

$$y = \frac{40}{20} = 2$$

When x increases by a factor of 4, y decreases by a factor of 4.

Final answer: $y = 2$

Graph.



The relationship plotted, with the data points marked.

Proportion 10 (cg4-010)

Difficulty: Foundation / Marks: 3

Given that x and y are inversely proportional, $y = 4$ when $x = 15$, and $y = a$ when $x = 3$, find a .

Worked solution.

Step 1: Find k .

$$k = xy = 4 \times 15 = 60$$

For inverse proportion, $xy = k$ is constant.

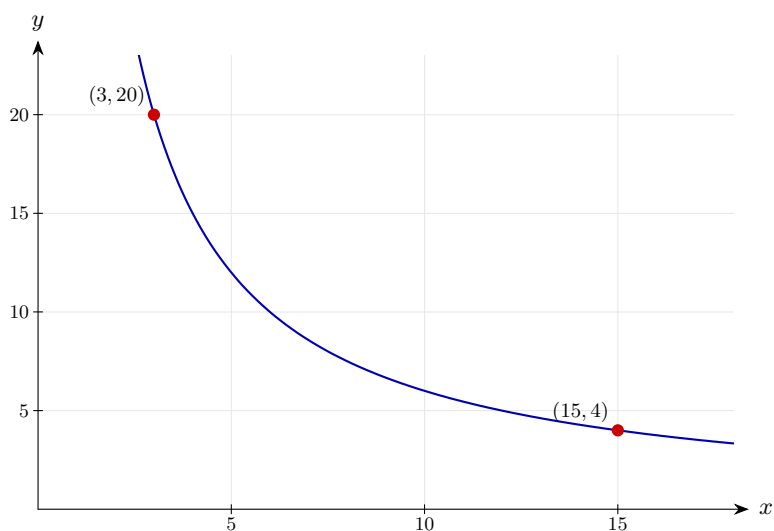
Step 2: Substitute $x = 3$.

$$a = \frac{60}{3} = 20$$

Divide the constant by the new x -value.

Final answer: $a = 20$

Graph.



The relationship plotted, with the data points marked.

Proportion 11 (cg4-011)

Difficulty: Foundation / Marks: 3

The time t hours to complete a journey is inversely proportional to the average speed v km/h. A journey takes 3 hours at 80 km/h. How long does it take at 60 km/h?

Worked solution.

Step 1: Write $t = \frac{k}{v}$ and find k .

$$k = t \times v = 3 \times 80 = 240$$

The constant k equals distance (240 km).

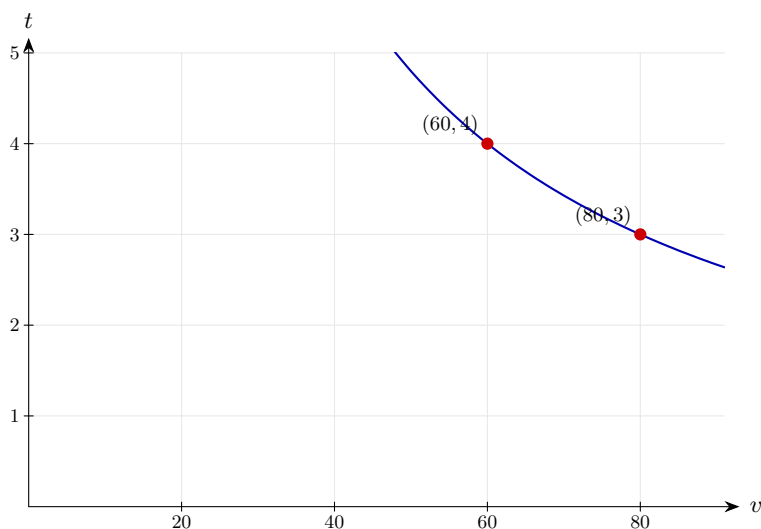
Step 2: Substitute $v = 60$.

$$t = \frac{240}{60} = 4 \text{ hours}$$

A lower speed means a longer journey time.

Final answer: 4 hours

Graph.



The relationship plotted, with the data points marked.

Proportion 12 (cg4-012)

Difficulty: Foundation / Marks: 3

Given that $y \propto \frac{1}{x}$, that $y = 9$ when $x = 4$, and that $y = 36$ when $x = a$, find a .

Worked solution.

Step 1: Find the constant $k = xy$.

$$k = 9 \times 4 = 36$$

For inverse proportion xy is always equal to k .

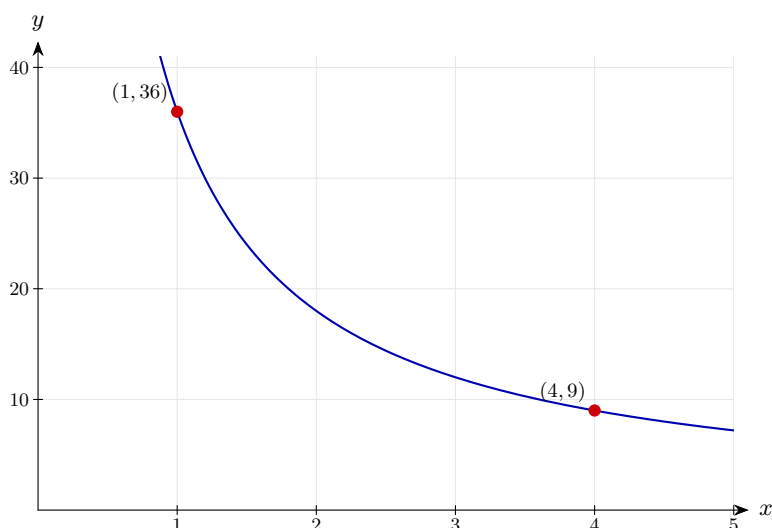
Step 2: Use the second pair to find a .

$$36 \times a = 36 \implies a = 1$$

Divide both sides by 36.

Final answer: $a = 1$

Graph.



The relationship plotted, with the data points marked.

Proportion 13 (cg4-013)

Difficulty: Foundation / Marks: 4

The number of days d needed to build a wall is inversely proportional to the number of workers w . With 6 workers the job takes 10 days.

a) How many days would 4 workers take?

b) How many workers are needed to complete the job in 5 days?

Worked solution.

Step 1: Find k .

$$k = d \times w = 10 \times 6 = 60$$

The constant represents the total worker-days needed.

Step 2: Part a: substitute $w = 4$.

$$d = \frac{60}{4} = 15 \text{ days}$$

Fewer workers means more days.

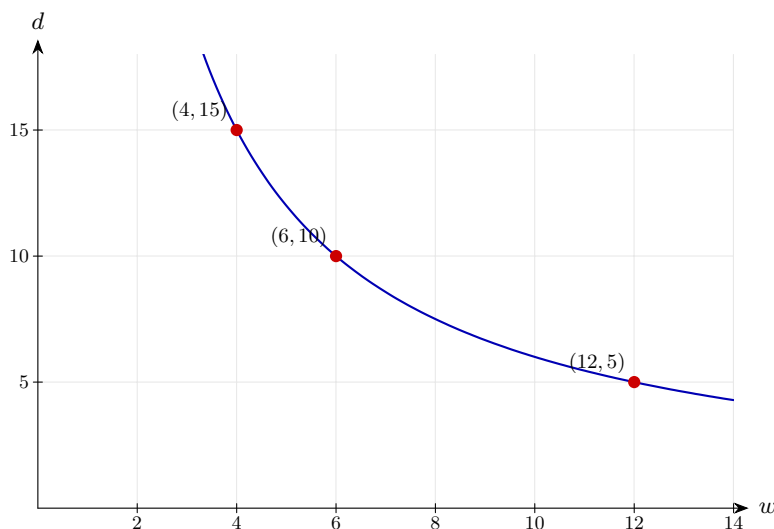
Step 3: Part b: substitute $d = 5$.

$$w = \frac{60}{5} = 12 \text{ workers}$$

More workers means fewer days.

Final answer: a) 15 days; b) 12 workers

Graph.



The relationship plotted, with the data points marked.

Proportion 14 (cg4-014)

Difficulty: Foundation / Marks: 4

Given that $y \propto \frac{1}{x}$, $y = a$ **when** $x = 6$, **and** $y = 6$ **when** $x = 24$ **(with** $a > 0$ **)**, **find** a .

Worked solution.

Step 1: Write $xy = k$ and use the first pair $y = a$, $x = 6$.

$$k = 6a$$

For inverse proportion the product xy is the same constant k for every pair. The first pair gives k in terms of the unknown a .

Step 2: Use the second pair $y = 6$, $x = 24$ to find a numerical value for k .

$$k = 24 \times 6 = 144$$

The second pair has no unknowns, so it pins down k exactly. Both expressions must equal the same k , so we can now equate them.

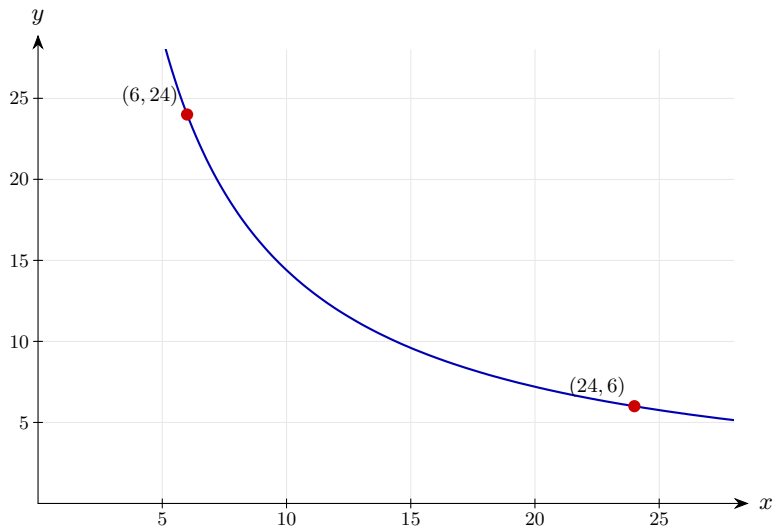
Step 3: Equate the two expressions for k and solve for a .

$$6a = 144 \implies a = 24$$

Divide both sides by 6. Check: when $x = 6$, $y = \frac{144}{6} = 24 = a$, as required.

Final answer: $a = 24$

Graph.



The relationship plotted, with the data points marked.

Proportion 15 (cg4-015)

Difficulty: Foundation / Marks: 3

The volume V of a fixed mass of gas is inversely proportional to the pressure P . When $P = 250$ Pa the volume is 4 m^3 . Find the pressure when the volume is 2.5 m^3 .

Worked solution.

Step 1: Write $V = \frac{k}{P}$ and find k .

$$k = VP = 4 \times 250 = 1000$$

k is the constant product of volume and pressure.

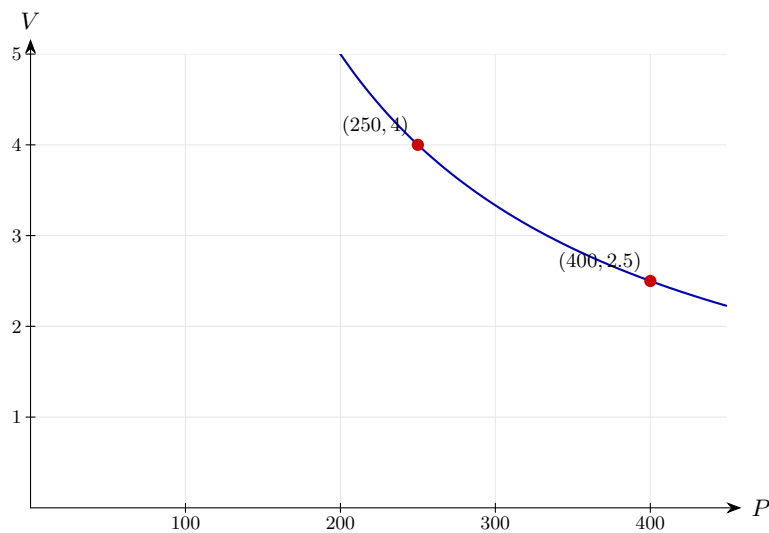
Step 2: Substitute $V = 2.5$.

$$P = \frac{1000}{2.5} = 400 \text{ Pa}$$

Halving the volume doubles the pressure (approximately).

Final answer: $P = 400 \text{ Pa}$

Graph.



The relationship plotted, with the data points marked.

Proportion 16 (cg4-016)

Difficulty: Foundation / Marks: 3

Given that $y \propto x^2$ and that $y = 75$ when $x = 5$, find y when $x = 4$.

Worked solution.

Step 1: Write $y = kx^2$ and substitute.

$$75 = k \times 25 \implies k = 3$$

Divide both sides by $5^2 = 25$.

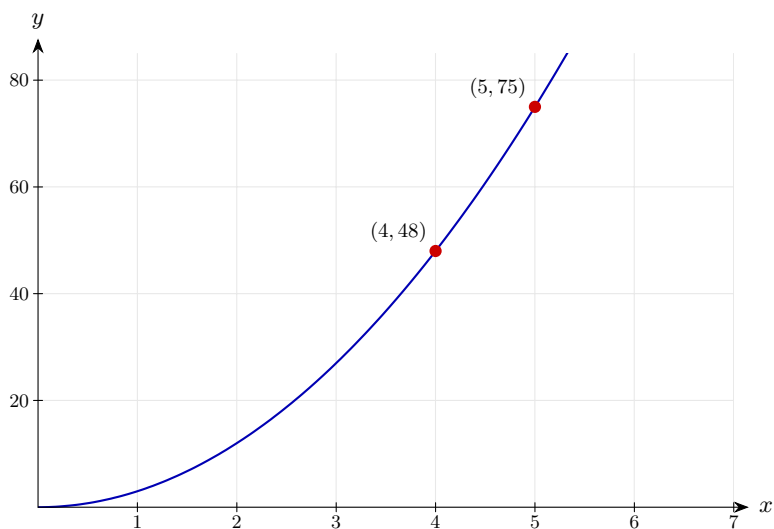
Step 2: Substitute $x = 4$.

$$y = 3 \times 16 = 48$$

Square x first, then multiply by k .

Final answer: $y = 48$

Graph.



The relationship plotted, with the data points marked.

Proportion 17 (cg4-017)

Difficulty: Foundation / Marks: 3

The surface area $A \text{ cm}^2$ of a sphere is directly proportional to the square of its radius $r \text{ cm}$. When $r = 3$, $A = 36\pi$. Find A when $r = 5$.

Worked solution.

Step 1: Write $A = kr^2$ and substitute $r = 3$, $A = 36\pi$.

$$36\pi = k \times 9 \implies k = 4\pi$$

Divide both sides by 9.

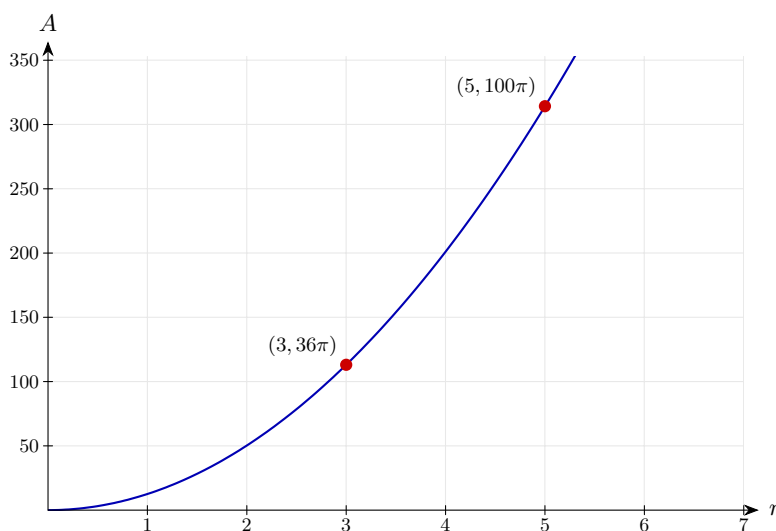
Step 2: Substitute $r = 5$.

$$A = 4\pi \times 25 = 100\pi \text{ cm}^2$$

The surface area of a sphere is indeed $4\pi r^2$.

Final answer: $A = 100\pi \text{ cm}^2$

Graph.



The relationship plotted, with the data points marked.

Proportion 18 (cg4-018)

Difficulty: Foundation / Marks: 4

Given that $y \propto x^2$ and $y = 20$ when $x = 2$, find the value of x (with $x > 0$) when $y = 125$.

Worked solution.

Step 1: Find k .

$$20 = k \times 4 \implies k = 5$$

Divide both sides by $2^2 = 4$.

Step 2: Substitute $y = 125$.

$$125 = 5x^2 \implies x^2 = 25$$

Divide both sides by 5.

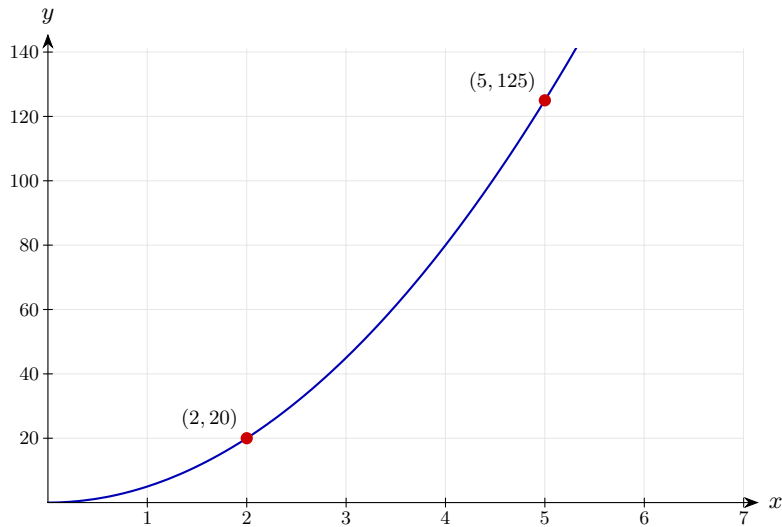
Step 3: Take the positive square root.

$$x = 5$$

Since $x > 0$, take the positive root only.

Final answer: $x = 5$

Graph.



The relationship plotted, with the data points marked.

Proportion 19 (cg4-019)

Difficulty: Foundation / *Marks: 6*

The braking distance d metres of a car is directly proportional to the square of its speed v km/h. At 30 km/h the braking distance is 9 m.

- Find the equation connecting d and v .
- Find the braking distance at 50 km/h.
- Find the speed at which the braking distance is 36 m.

Worked solution.

Step 1: Write $d = kv^2$ and find k .

$$9 = k \times 900 \implies k = \frac{9}{900} = \frac{1}{100}$$

Substitute $d = 9, v = 30$ and divide.

Step 2: Part a: state the equation.

$$d = \frac{v^2}{100}$$

This is the proportionality equation.

Step 3: Part b: substitute $v = 50$.

$$d = \frac{2500}{100} = 25 \text{ m}$$

The braking distance is 25 m at 50 km/h.

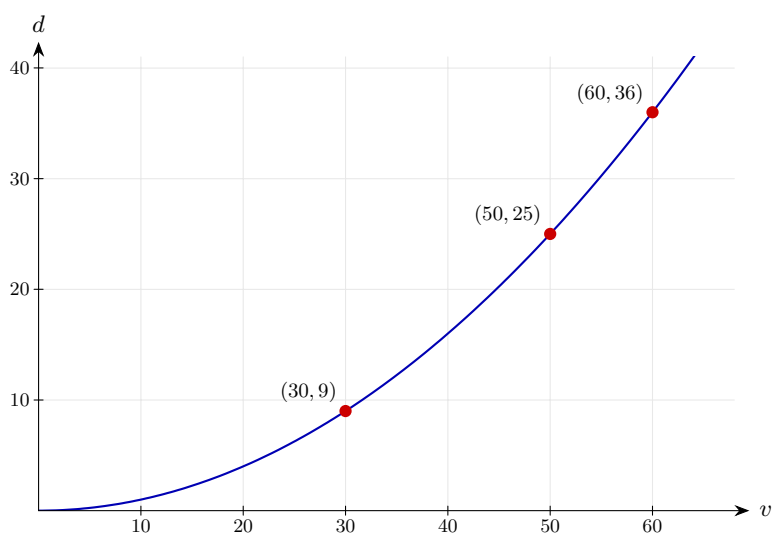
Step 4: Part c: substitute $d = 36$ and solve.

$$36 = \frac{v^2}{100} \implies v^2 = 3600 \implies v = 60 \text{ km/h}$$

Take the positive square root (speed is positive).

Final answer: a) $d = \frac{v^2}{100}$; b) 25 m; c) 60 km/h

Graph.



The relationship plotted, with the data points marked.

Proportion 20 (cg4-020)

Difficulty: Foundation / Marks: 4

Given that $y \propto x^2$ and $y = 12$ when $x = 2$, sketch the graph of y against x for $x \geq 0$ and find y when $x = 3$.

Worked solution.

Step 1: Find k .

$$12 = k \times 4 \implies k = 3$$

Substitute the known values.

Step 2: State the equation.

$$y = 3x^2$$

This is a parabola through the origin.

Step 3: Describe the sketch.

Graph: parabola through (0,0), (1,3), (2,12), (3,27)

The curve starts at the origin, opens upward, passes through the points listed.

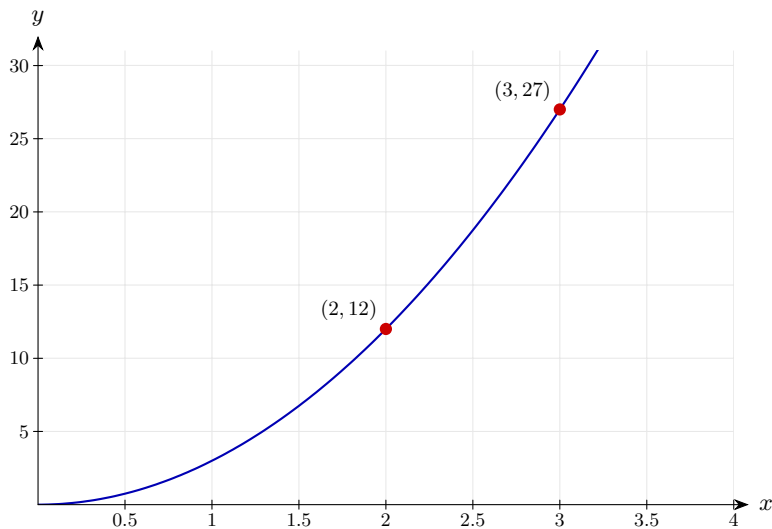
Step 4: Find y when $x = 3$.

$$y = 3 \times 9 = 27$$

Square x then multiply by k .

Final answer: $y = 27$ when $x = 3$; graph is a parabola through the origin.

Graph.



The relationship plotted, with the data points marked.

Proportion 21 (cg4-021)

Difficulty: Foundation / Marks: 3

The power P watts dissipated in a resistor is directly proportional to the square of the current I amps. When $I = 3$ A, $P = 45$ W. Find P when $I = 5$ A.

Worked solution.

Step 1: Write $P = kI^2$ and find k .

$$45 = k \times 9 \implies k = 5$$

$3^2 = 9$; divide both sides by 9.

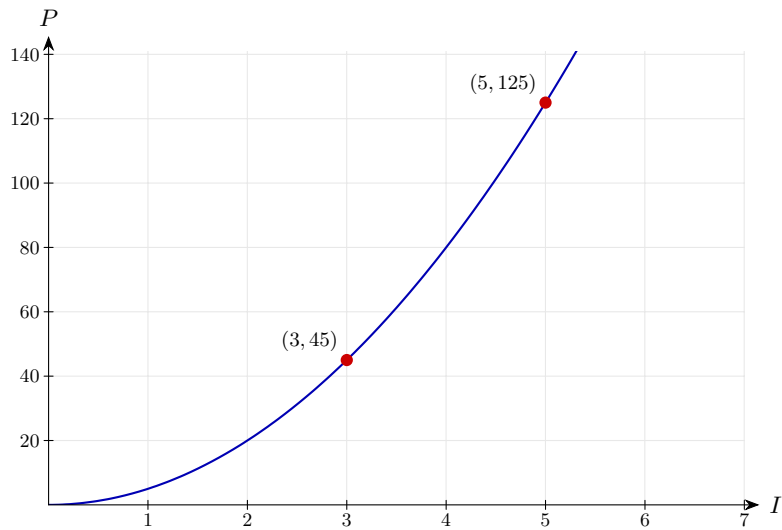
Step 2: Substitute $I = 5$.

$$P = 5 \times 25 = 125 \text{ W}$$

The power dissipated is 125 W.

Final answer: $P = 125$ W

Graph.



The relationship plotted, with the data points marked.

Proportion 22 (cg4-022)

Difficulty: Foundation / Marks: 3

Given that $y \propto \frac{1}{x^2}$ and that $y = 16$ when $x = 3$, find y when $x = 6$.

Worked solution.

Step 1: Write $y = \frac{k}{x^2}$ and find k .

$$16 = \frac{k}{9} \implies k = 144$$

Multiply both sides by $3^2 = 9$.

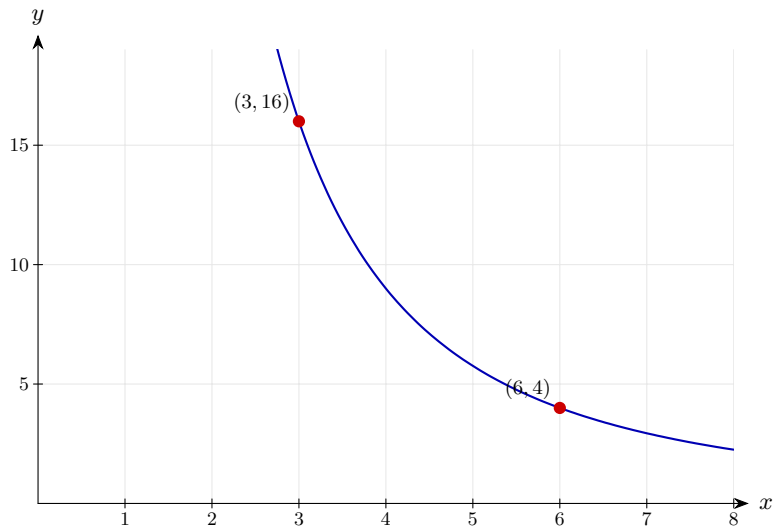
Step 2: Substitute $x = 6$.

$$y = \frac{144}{36} = 4$$

When x doubles, y decreases by a factor of 4.

Final answer: $y = 4$

Graph.



The relationship plotted, with the data points marked.

Proportion 23 (cg4-023)

Difficulty: Foundation / Marks: 3

The intensity I of light from a lamp is inversely proportional to the square of the distance d metres from the lamp. At $d = 2$ m, $I = 50$ lux. Find I when $d = 5$ m.

Worked solution.

Step 1: Write $I = \frac{k}{d^2}$ and find k .

$$50 = \frac{k}{4} \implies k = 200$$

Multiply both sides by $2^2 = 4$.

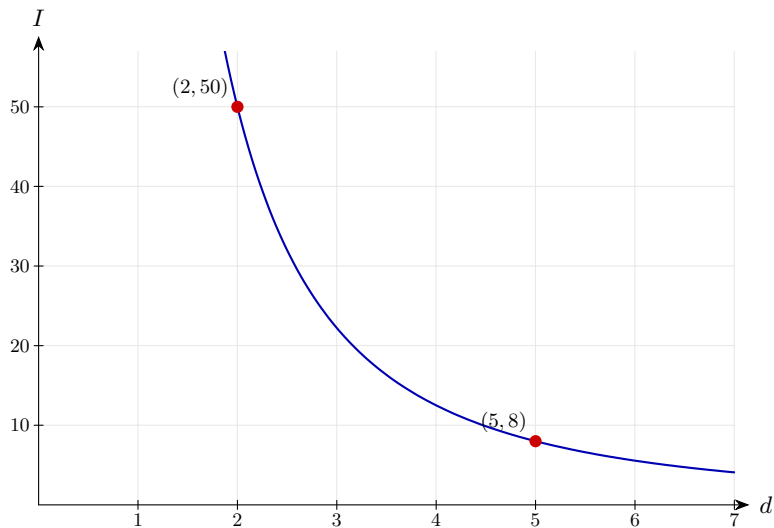
Step 2: Substitute $d = 5$.

$$I = \frac{200}{25} = 8 \text{ lux}$$

The intensity decreases sharply as distance increases.

Final answer: $I = 8$ lux

Graph.



The relationship plotted, with the data points marked.

Proportion 24 (cg4-024)

Difficulty: Foundation / Marks: 4

Given that $y \propto \frac{1}{x^2}$, $y = 25$ when $x = 2$, find the positive value of x when $y = 1$.

Worked solution.

Step 1: Find k .

$$k = 25 \times 4 = 100$$

$$k = y \cdot x^2 = 25 \times 2^2 = 100.$$

Step 2: Substitute $y = 1$.

$$1 = \frac{100}{x^2} \implies x^2 = 100$$

Multiply both sides by x^2 and rearrange.

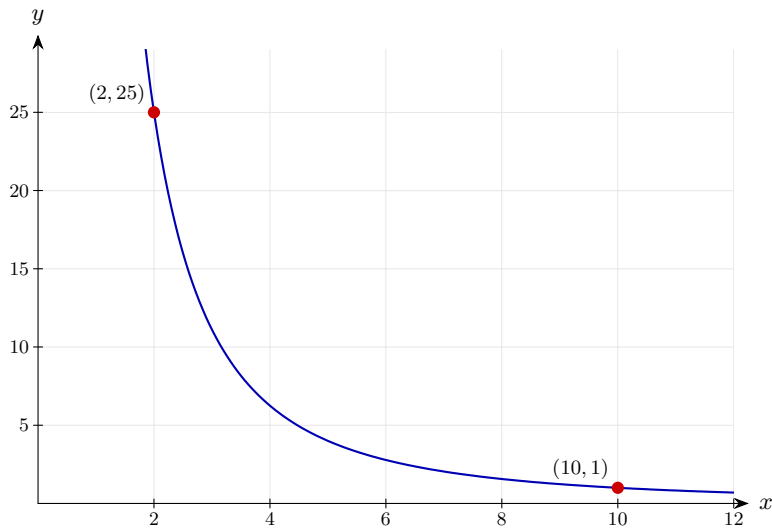
Step 3: Take the positive square root.

$$x = 10$$

Since $x > 0$ we take the positive root.

Final answer: $x = 10$

Graph.



The relationship plotted, with the data points marked.

Proportion 25 (cg4-025)

Difficulty: Foundation / Marks: 5

The gravitational force F N between two objects is inversely proportional to the square of the distance r m between them. When $r = 3$ m, $F = 20$ N. Find the force when $r = 6$ m, and hence state the effect of tripling the original distance on the force.

Worked solution.

Step 1: Write $F = \frac{k}{r^2}$ and find k .

$$20 = \frac{k}{9} \implies k = 180$$

Multiply both sides by 9.

Step 2: Find F when $r = 6$.

$$F = \frac{180}{36} = 5 \text{ N}$$

The force is 5 N when the distance is 6 m.

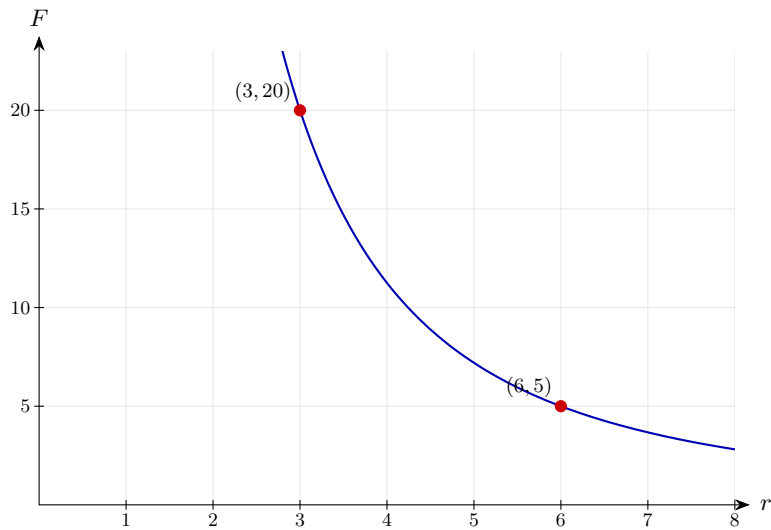
Step 3: State the effect of tripling the distance ($r \rightarrow 3r$).

$$F_{\text{new}} = \frac{k}{(3r)^2} = \frac{k}{9r^2} = \frac{F_{\text{old}}}{9}$$

Tripling the distance reduces the force by a factor of 9.

Final answer: $F = 5$ N when $r = 6$ m; tripling the distance reduces the force to $\frac{1}{9}$ of its original value.

Graph.



The relationship plotted, with the data points marked.

Proportion 26 (cg4-026)

Difficulty: Foundation / Marks: 3

Given that $y \propto \frac{1}{x^2}$, $y = 9$ when $x = 4$, and $y = a$ when $x = 12$, find a .

Worked solution.

Step 1: Find k .

$$k = 9 \times 16 = 144$$

$$k = y \cdot x^2 = 9 \times 4^2.$$

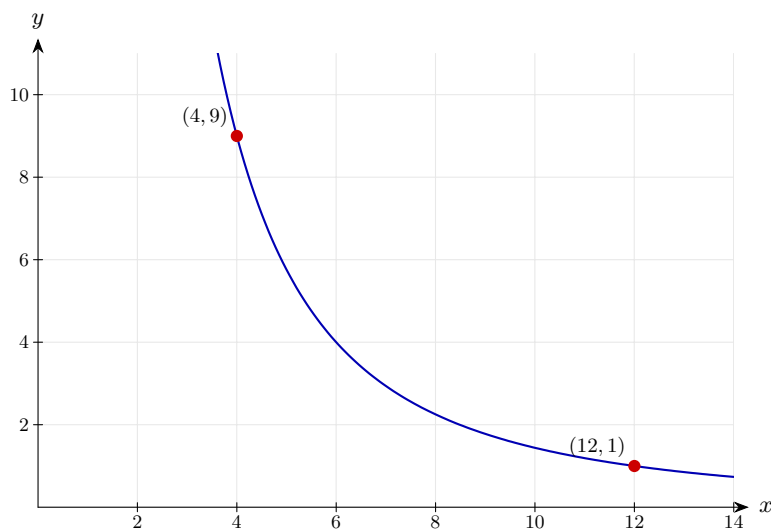
Step 2: Substitute $x = 12$.

$$a = \frac{144}{144} = 1$$

$$12^2 = 144, \text{ so } a = \frac{144}{144} = 1.$$

Final answer: $a = 1$

Graph.



The relationship plotted, with the data points marked.

Proportion 27 (cg4-027)

Difficulty: Foundation / Marks: 3

Given that $y \propto \sqrt{x}$ and that $y = 15$ when $x = 25$, find y when $x = 64$.

Worked solution.

Step 1: Write $y = k\sqrt{x}$ and find k .

$$15 = k\sqrt{25} = 5k \implies k = 3$$

$\sqrt{25} = 5$; divide both sides by 5.

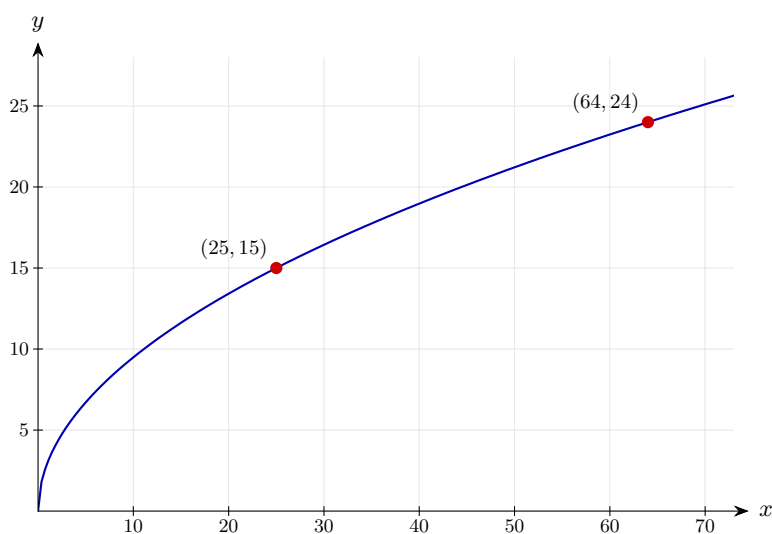
Step 2: Substitute $x = 64$.

$$y = 3\sqrt{64} = 3 \times 8 = 24$$

$\sqrt{64} = 8$.

Final answer: $y = 24$

Graph.



The relationship plotted, with the data points marked.

Proportion 28 (cg4-028)

Difficulty: Foundation / Marks: 4

Given that $y \propto \sqrt{x}$ and $y = 6$ when $x = 9$, find the exact value of x when $y = 10$.

Worked solution.

Step 1: Find k .

$$6 = k\sqrt{9} = 3k \implies k = 2$$

Divide both sides by $\sqrt{9} = 3$.

Step 2: Substitute $y = 10$.

$$10 = 2\sqrt{x} \implies \sqrt{x} = 5$$

Divide both sides by 2.

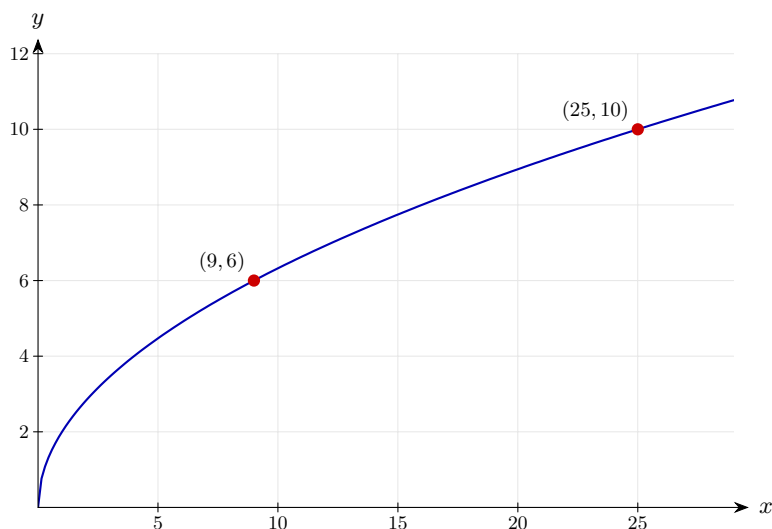
Step 3: Square both sides.

$$x = 25$$

$$(\sqrt{x})^2 = x.$$

Final answer: $x = 25$

Graph.



The relationship plotted, with the data points marked.

Proportion 29 (cg4-029)

Difficulty: Foundation / Marks: 4

Prove that if $y \propto x$ and $z \propto y^2$, then $z \propto x^2$.

Worked solution.

Step 1: Write equations for each proportionality.

$$y = k_1x \quad \text{and} \quad z = k_2y^2$$

Replace each $\propto$ with $= k \cdot$.

Step 2: Substitute the expression for y into the equation for z .

$$z = k_2(k_1x)^2 = k_2k_1^2x^2$$

Expand the bracket: $(k_1x)^2 = k_1^2x^2$.

Step 3: Identify the combined constant.

$$z = Kx^2 \quad \text{where } K = k_1^2k_2 \text{ is a constant}$$

Since K is a product of constants it is itself a constant, so $z \propto x^2$. $\square$

Final answer: Proved: $z \propto x^2$

Proportion 30 (cg4-030)

Difficulty: Foundation / Marks: 4

Prove that if $y \propto \frac{1}{x}$ and $z \propto \frac{1}{y}$, then $z \propto x$.

Worked solution.

Step 1: Write equations for each relationship.

$$y = \frac{k_1}{x} \quad \text{and} \quad z = \frac{k_2}{y}$$

Express each proportionality as an equation.

Step 2: Substitute y into the expression for z .

$$z = \frac{k_2}{\frac{k_1}{x}} = \frac{k_2 x}{k_1}$$

Dividing by a fraction is multiplying by its reciprocal.

Step 3: Write in the form $z = Kx$.

$$z = \frac{k_2}{k_1} \cdot x = Kx \quad \text{where } K = \frac{k_2}{k_1}$$

Since K is a constant, $z \propto x$. $\square$

Final answer: Proved: $z \propto x$

Proportion 31 (cg4-031)

Difficulty: Foundation / Marks: 4

The period T seconds of a pendulum is directly proportional to the square root of its length L cm. A pendulum of length 100 cm has a period of 2 s. Find the length needed to give a period of 3 s.

Worked solution.

Step 1: Write $T = k\sqrt{L}$ and find k .

$$2 = k\sqrt{100} = 10k \implies k = 0.2$$

$\sqrt{100} = 10$; divide both sides by 10.

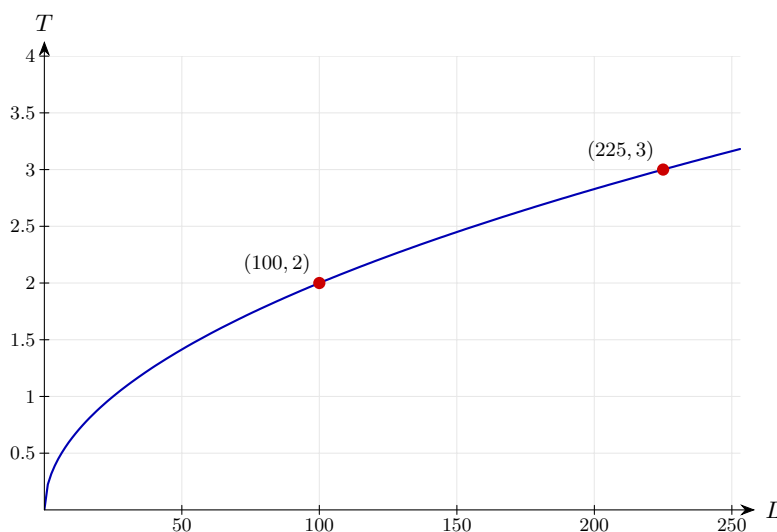
Step 2: Substitute $T = 3$ and solve for L .

$$3 = 0.2\sqrt{L} \implies \sqrt{L} = 15 \implies L = 225 \text{ cm}$$

Divide by 0.2 then square both sides.

Final answer: $L = 225$ cm

Graph.



The relationship plotted, with the data points marked.

Proportion 32 (cg4-032)

Difficulty: Foundation / Marks: 8

The resistance R ohms of a wire is directly proportional to its length l m and inversely proportional to the square of its diameter d mm, so $R \propto \frac{l}{d^2}$. A wire of length 5 m and diameter 2 mm has resistance 10Ω .

- Find the equation connecting R , l , and d .
- Find the resistance of a wire of length 20 m and diameter 4 mm.
- A wire has length 3 m, resistance 7.5Ω . Find its diameter.

Worked solution.

Step 1: Write $R = \frac{kl}{d^2}$ and substitute the known values.

$$10 = \frac{k \times 5}{4} \implies 10 = \frac{5k}{4} \implies k = 8$$

$d^2 = 4$; multiply both sides by 4, then divide by 5.

Step 2: Part a: state the equation.

$$R = \frac{8l}{d^2}$$

This combines direct and inverse proportion.

Step 3: Part b: substitute $l = 20, d = 4$.

$$R = \frac{8 \times 20}{16} = \frac{160}{16} = 10 \Omega$$

Doubling length doubles R ; doubling diameter quarters R — net effect: unchanged.

Step 4: Part c: substitute $l = 3$, $R = 7.5$ and solve for d .

$$7.5 = \frac{8 \times 3}{d^2} = \frac{24}{d^2} \implies d^2 = \frac{24}{7.5} = 3.2 \implies d = \sqrt{3.2} = \frac{4\sqrt{5}}{5} \approx 1.79 \text{ mm}$$

Take the positive square root.

Final answer: a) $R = \frac{8l}{d^2}$; b) 10Ω ; c) $d = \sqrt{3.2} \approx 1.79 \text{ mm}$

Proportion 33 (cg4-033)

Difficulty: Foundation | Marks: 7

The variable y is directly proportional to x^2 . When $x = 2$, $y = 28$.

a) Find y when $x = 5$.

b) Find the exact values of x when $y = 63$.

c) Show that when x is tripled, y increases by a factor of 9.

Worked solution.

Step 1: Find k .

$$28 = k \times 4 \implies k = 7$$

So $y = 7x^2$.

Step 2: Part a: substitute $x = 5$.

$$y = 7 \times 25 = 175$$

$$5^2 = 25.$$

Step 3: Part b: substitute $y = 63$.

$$63 = 7x^2 \implies x^2 = 9 \implies x = \pm 3$$

Both positive and negative roots are valid.

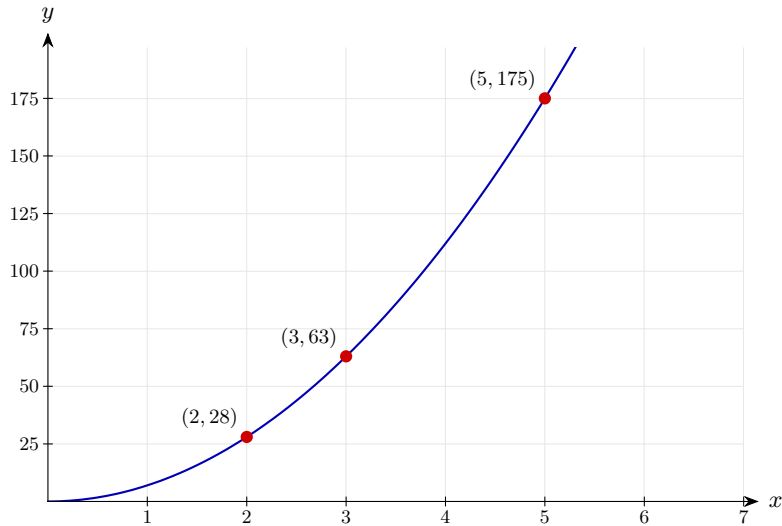
Step 4: Part c: replace x with $3x$.

$$y_{\text{new}} = 7(3x)^2 = 7 \times 9x^2 = 9 \times 7x^2 = 9y$$

$(3x)^2 = 9x^2$, so y is multiplied by 9. $\square$

Final answer: a) $y = 175$; b) $x = \pm 3$; c) shown

Graph.



The relationship plotted, with the data points marked.

Proportion 34 (cg4-034)

Difficulty: Foundation / Marks: 8

The variable P is inversely proportional to Q . When $Q = 5$, $P = 12$. The variable R is directly proportional to Q^2 . When $Q = 2$, $R = 20$.

- Express P in terms of Q .
- Express R in terms of Q .
- Find the value of Q for which $P = R$.

Worked solution.

Step 1: Part a: find k_1 from $P = \frac{k_1}{Q}$.

$$12 = \frac{k_1}{5} \implies k_1 = 60 \implies P = \frac{60}{Q}$$

Multiply both sides by 5.

Step 2: Part b: find k_2 from $R = k_2 Q^2$.

$$20 = k_2 \times 4 \implies k_2 = 5 \implies R = 5Q^2$$

Divide both sides by $2^2 = 4$.

Step 3: Part c: set $P = R$ and solve.

$$\frac{60}{Q} = 5Q^2 \implies 60 = 5Q^3 \implies Q^3 = 12$$

Multiply both sides by Q (assuming $Q \neq 0$), then divide by 5.

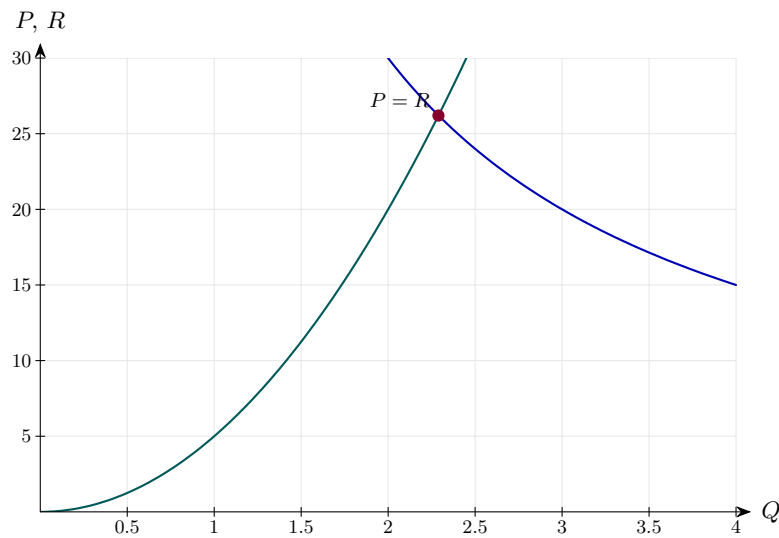
Step 4: Find the exact value of Q .

$$Q = \sqrt[3]{12}$$

Take the cube root of both sides. $Q = \sqrt[3]{12} \approx 2.29$.

Final answer: a) $P = \frac{60}{Q}$; b) $R = 5Q^2$; c) $Q = \sqrt[3]{12}$

Graph.



The relationship plotted, with the data points marked.

Proportion 35 (cg4-035)

Difficulty: Foundation / Marks: 10

The variable y is inversely proportional to the square of x . When $x = 2$, $y = 50$.

- Find the value of y when $x = 10$.
- Find the exact values of x when $y = 8$.
- Show that when x is halved, y increases by a factor of 4.
- Given that $z \propto y$ and $z = 30$ when $y = 50$, find z when $x = 5$.

Worked solution.

Step 1: Find k : write $y = \frac{k}{x^2}$.

$$50 = \frac{k}{4} \implies k = 200$$

Multiply both sides by $2^2 = 4$.

Step 2: Part a: substitute $x = 10$.

$$y = \frac{200}{100} = 2$$

$$10^2 = 100.$$

Step 3: Part b: substitute $y = 8$ and solve.

$$8 = \frac{200}{x^2} \implies x^2 = 25 \implies x = \pm 5$$

Divide 200 by 8 to get $x^2 = 25$, then take square roots.

Step 4: Part c: replace x with $\frac{x}{2}$.

$$y_{\text{new}} = \frac{200}{\left(\frac{x}{2}\right)^2} = \frac{200}{\frac{x^2}{4}} = \frac{800}{x^2} = 4 \times \frac{200}{x^2} = 4y$$

Halving x quarters x^2 , which quadruples y . $\square$

Step 5: Part d: find m from $z = my$.

$$30 = m \times 50 \implies m = \frac{3}{5}$$

Divide both sides by 50.

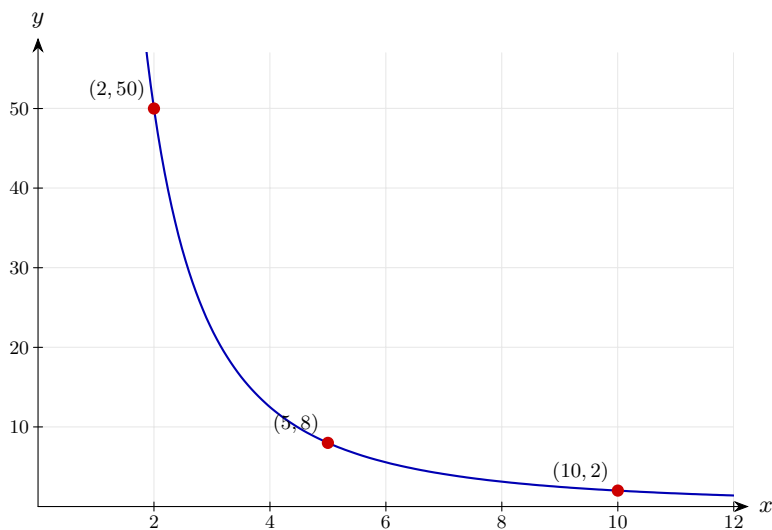
Step 6: Find y when $x = 5$, then find z .

$$y = \frac{200}{25} = 8 \implies z = \frac{3}{5} \times 8 = \frac{24}{5} = 4.8$$

$5^2 = 25$; multiply by m .

Final answer: a) $y = 2$; b) $x = \pm 5$; c) shown; d) $z = 4.8$

Graph.



The relationship plotted, with the data points marked.

Proportion 36 (cg4-036)

Difficulty: Foundation / Marks: 3

Given that y is directly proportional to x and $y = 15$ when $x = 5$, find y when $x = 8$.

Worked solution.

Step 1: Find k

$$y = kx \implies 15 = 5k \implies k = 3$$

$y \propto x$ means $y = kx$ for some constant k . Substitute the given pair and divide both sides by 5 to isolate k .

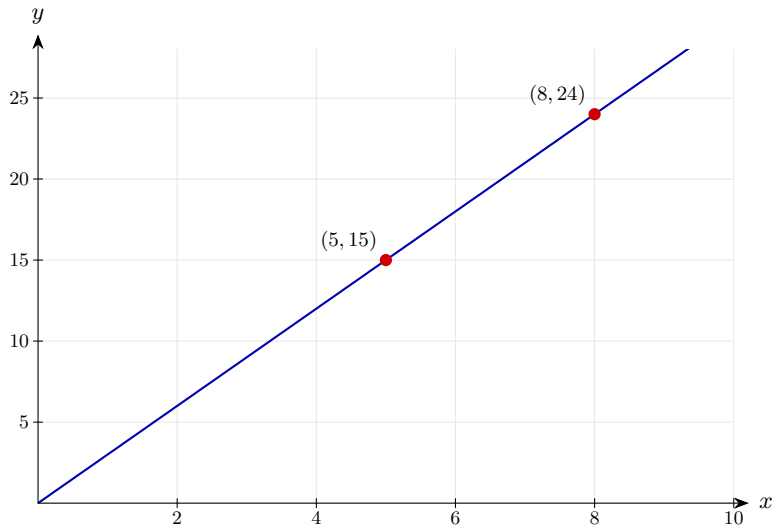
Step 2: Find y

$$y = 3 \times 8 = 24$$

Now that k is known, the equation $y = 3x$ works for any x ; substitute the new value.

Final answer: $y = 24$

Graph.



The relationship plotted, with the data points marked.

Proportion 37 (cg4-037)

Difficulty: Foundation / Marks: 3

Given that y is inversely proportional to x and $y = 6$ when $x = 4$, find y when $x = 12$.

Worked solution.

Step 1: Find k

$$y = \frac{k}{x} \implies 6 = \frac{k}{4} \implies k = 24$$

$y \propto \frac{1}{x}$ translates to $y = \frac{k}{x}$. Multiply both sides by 4 to clear the fraction and isolate k .

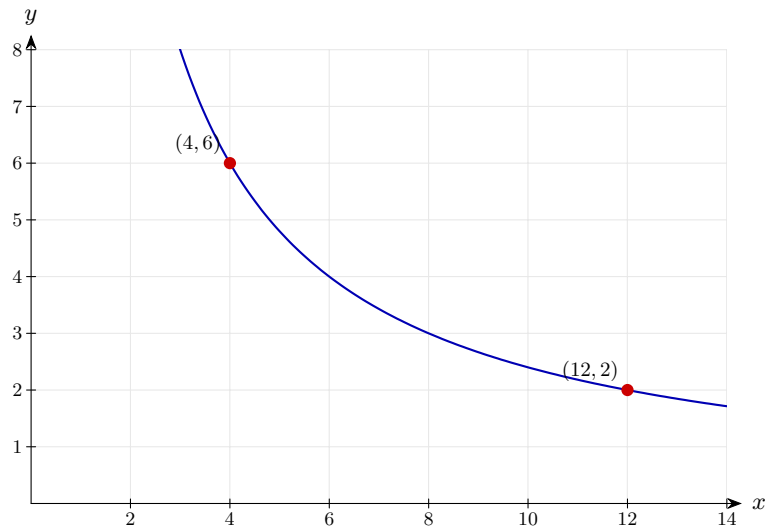
Step 2: Find y

$$y = \frac{24}{12} = 2$$

Sense-check: tripling x (from 4 to 12) should cut y to one third ($6 \rightarrow 2$), and it does.

Final answer: $y = 2$

Graph.



The relationship plotted, with the data points marked.

Proportion 38 (cg4-038)

Difficulty: Foundation / Marks: 3

y is proportional to x^2 . When $x = 3$, $y = 36$. Find y when $x = 5$.

Worked solution.

Step 1: Find k

$$y = kx^2 \implies 36 = 9k \implies k = 4$$

$y \propto x^2$ gives $y = kx^2$. Square $x = 3$ first to get 9, then divide. A common slip is to write $36 = 3k$ — remember it is x^2 , not x .

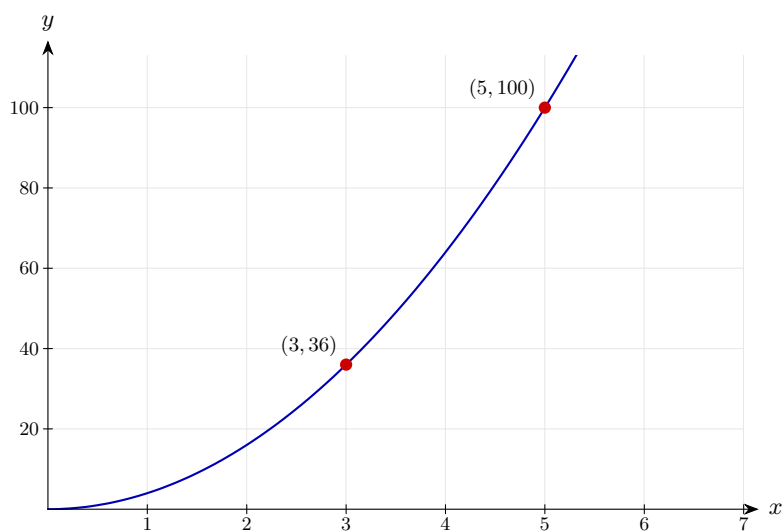
Step 2: Find y

$$y = 4 \times 25 = 100$$

Square $x = 5$ first (giving 25), then multiply by k . The order matters because $kx^2 \neq (kx)^2$.

Final answer: $y = 100$

Graph.



The relationship plotted, with the data points marked.

Proportion 39 (cg4-039)

Difficulty: Foundation / Marks: 3

y is inversely proportional to $\sqrt{x}$. When $x = 9$, $y = 4$. Find y when $x = 16$.

Worked solution.

Step 1: Find k

$$y = \frac{k}{\sqrt{x}} \implies 4 = \frac{k}{3} \implies k = 12$$

$y \propto \frac{1}{\sqrt{x}}$ means $y = \frac{k}{\sqrt{x}}$. Evaluate $\sqrt{9} = 3$ first, then multiply both sides by 3.

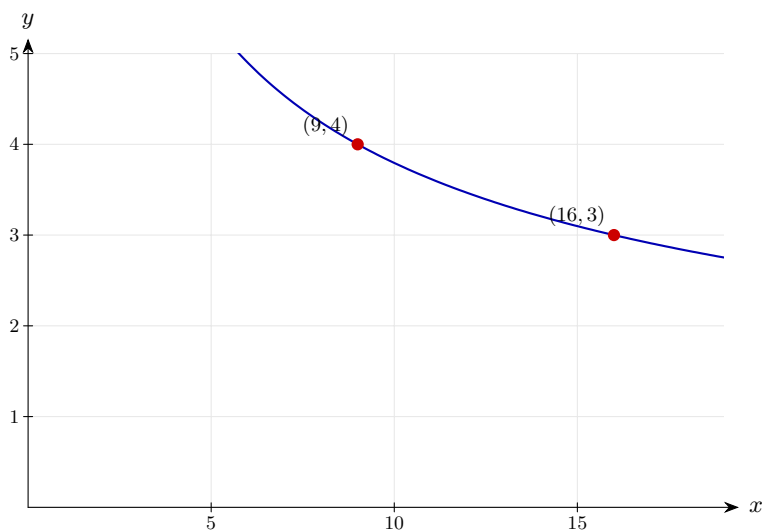
Step 2: Find y

$$y = \frac{12}{\sqrt{16}} = \frac{12}{4} = 3$$

Take the square root before dividing — $\sqrt{16} = 4$, not 16. Pitfall: students often forget the root and divide by x instead.

Final answer: $y = 3$

Graph.



The relationship plotted, with the data points marked.

Proportion 40 (cg4-040)

Difficulty: Foundation / Marks: 3

The cost C of fuel is directly proportional to the volume V litres bought. If 25 litres costs £37.50, find the cost of 40 litres.

Worked solution.

Step 1: Find k

$$C = kV \implies 37.50 = 25k \implies k = 1.50$$

Direct proportion gives $C = kV$. Here k is the price per litre — divide total cost by litres to find it.

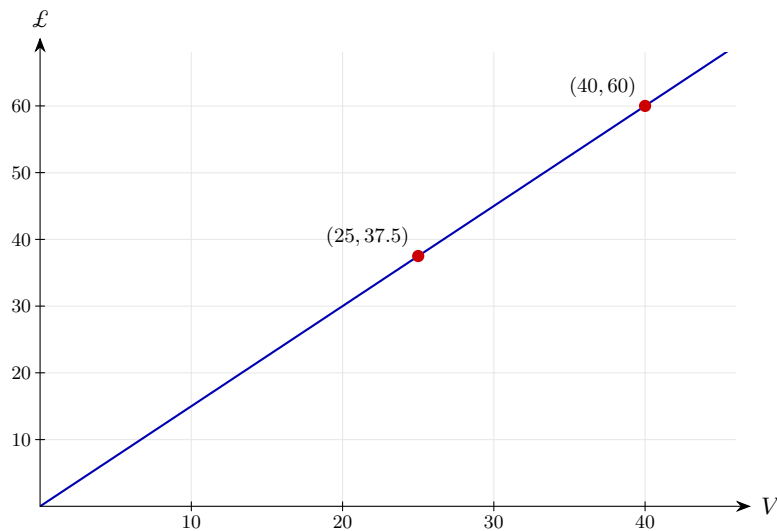
Step 2: Find C

$$C = 1.50 \times 40 = 60$$

Multiply the unit price by the new volume. Always sense-check: 40 litres is more than 25, so the cost should be more than £37.50.

Final answer: £60

Graph.



The relationship plotted, with the data points marked.

Proportion 41 (cg4-041)

Difficulty: Foundation / Marks: 4

y is proportional to x^3 . When $x = 2$, $y = 40$. Find x when $y = 135$.

Worked solution.

Step 1: Find k

$$y = kx^3 \implies 40 = 8k \implies k = 5$$

$y \propto x^3$ gives $y = kx^3$. Cube $x = 2$ to get 8 before dividing — do not confuse x^3 with $3x$.

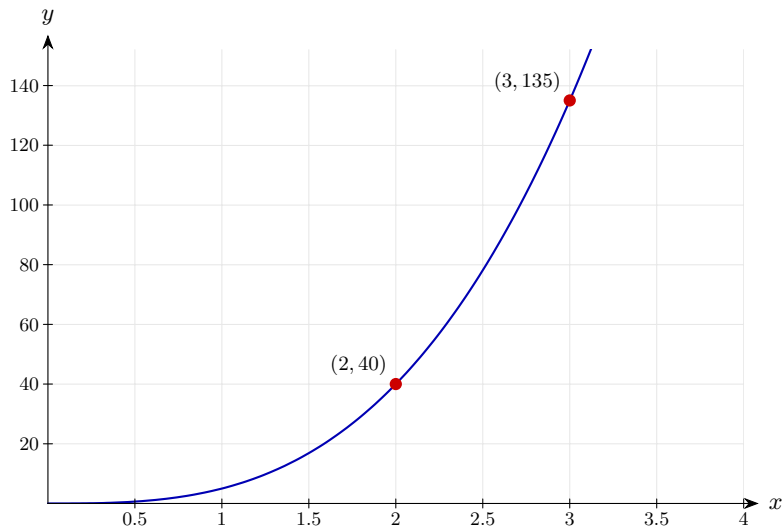
Step 2: Find x

$$135 = 5x^3 \implies x^3 = 27 \implies x = 3$$

Divide both sides by 5 to isolate x^3 , then take the cube root. Unlike square roots, cube roots have a single real value, so no $\pm$ is needed.

Final answer: $x = 3$

Graph.



The relationship plotted, with the data points marked.

Proportion 42 (cg4-042)

Difficulty: Foundation / Marks: 3

The time T to travel a fixed distance is inversely proportional to speed s . At 60 km/h the journey takes 2 hours. How long at 80 km/h?

Worked solution.

Step 1: Find k

$$T = \frac{k}{s} \implies 2 = \frac{k}{60} \implies k = 120$$

Inverse proportion gives $T = \frac{k}{s}$. Here k is the fixed distance in km (since distance = speed $\times$ time). Multiply both sides by 60.

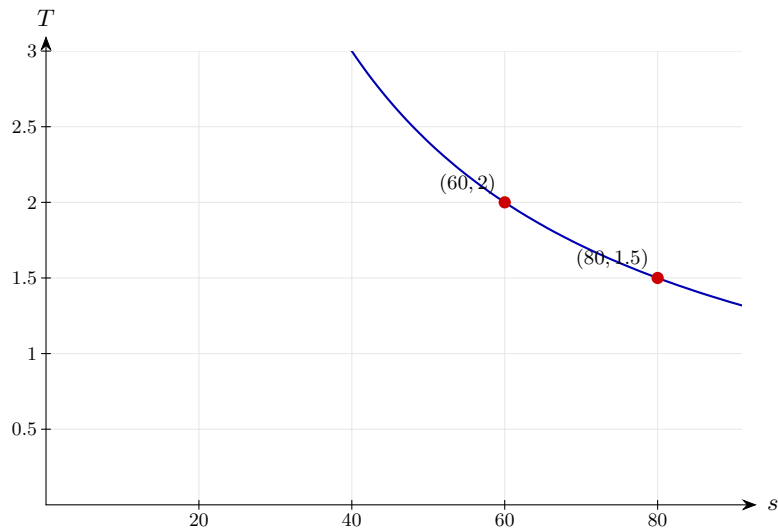
Step 2: Find T

$$T = \frac{120}{80} = 1.5 \text{ hours}$$

Sense-check: a higher speed (80 vs 60) should give a shorter time, and $1.5 < 2$ hours, so this is consistent.

Final answer: 1.5 hours (1 hour 30 minutes)

Graph.



The relationship plotted, with the data points marked.

Proportion 43 (cg4-043)

Difficulty: Foundation / Marks: 4

P is proportional to $\sqrt{Q}$. When $Q = 25$, $P = 10$. Find Q when $P = 14$.

Worked solution.

Step 1: Find k

$$P = k\sqrt{Q} \implies 10 = 5k \implies k = 2$$

$P \propto \sqrt{Q}$ translates to $P = k\sqrt{Q}$. Take $\sqrt{25} = 5$ first, then divide both sides by 5.

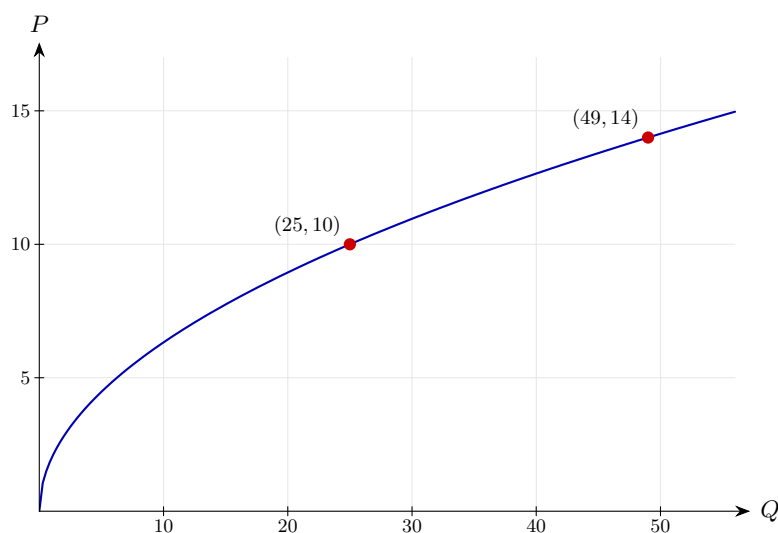
Step 2: Find Q

$$14 = 2\sqrt{Q} \implies \sqrt{Q} = 7 \implies Q = 49$$

Divide by 2 to isolate $\sqrt{Q}$, then square both sides to get Q . Squaring last avoids working with Q under a root.

Final answer: $Q = 49$

Graph.



The relationship plotted, with the data points marked.

Proportion 44 (cg4-044)

Difficulty: Foundation / Marks: 3

y is inversely proportional to x^2 . When $x = 2$, $y = 5$. Find y when $x = 10$.

Worked solution.

Step 1: Find k

$$y = \frac{k}{x^2} \implies 5 = \frac{k}{4} \implies k = 20$$

$y \propto \frac{1}{x^2}$ means $y = \frac{k}{x^2}$. Square $x = 2$ to get 4, then multiply both sides by 4 to clear the fraction.

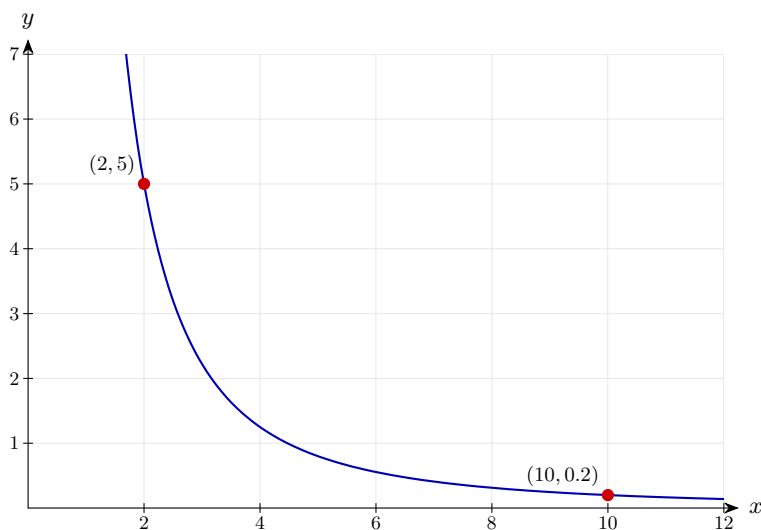
Step 2: Find y

$$y = \frac{20}{100} = 0.2$$

x increased 5-fold, so x^2 grew 25-fold and y shrank by a factor of 25 (from 5 to 0.2). That matches the inverse-square law.

Final answer: $y = 0.2$

Graph.



The relationship plotted, with the data points marked.

Proportion 45 (cg4-045)

Difficulty: Foundation / Marks: 3

If x is doubled, what happens to y when: (a) $y \propto x$; (b) $y \propto x^2$; (c) $y \propto \frac{1}{x}$?

Worked solution.

Step 1: (a) $y = kx$

$$y_{\text{new}} = k(2x) = 2kx = 2y$$

Replace x with $2x$ in $y = kx$. The factor of 2 pulls out directly, so the new value is exactly twice the old — y doubles.

Step 2: (b) $y = kx^2$

$$y_{\text{new}} = k(2x)^2 = 4kx^2 = 4y$$

Crucially $(2x)^2 = 4x^2$, not $2x^2$ — the bracket squares both the 2 and the x . So y is multiplied by 4.

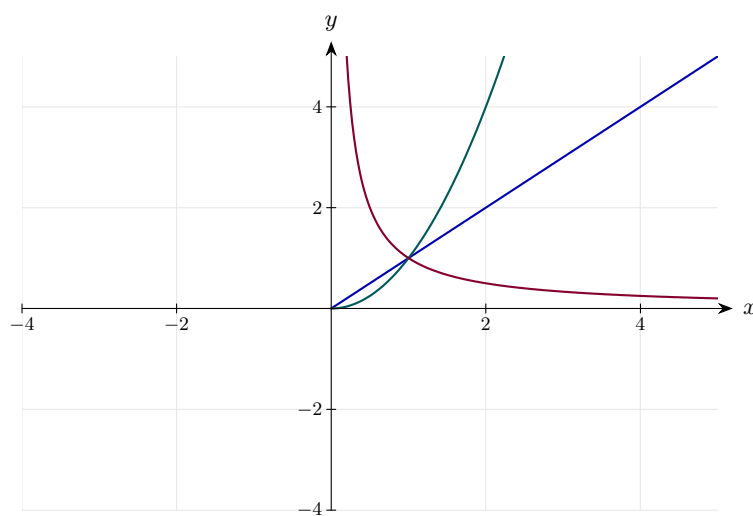
Step 3: (c) $y = k/x$

$$y_{\text{new}} = \frac{k}{2x} = \frac{y}{2}$$

The 2 ends up on the denominator, so y is divided by 2. Doubling x halves y for inverse proportion.

Final answer: (a) doubles; (b) quadruples; (c) halves

Graph.



The relationship plotted, with the data points marked.

Proportion 46 (cg4-046)

Difficulty: Foundation / Marks: 3

The weight W of a sphere is proportional to the cube of its radius r . A sphere of radius 3 cm weighs 108 g. Find the weight of a sphere of radius 5 cm.

Worked solution.

Step 1: Find k

$$W = kr^3 \implies 108 = 27k \implies k = 4$$

$W \propto r^3$ gives $W = kr^3$. Cube $r = 3$ to get 27 before dividing — that comes from $3 \times 3 \times 3$, not 3×3 .

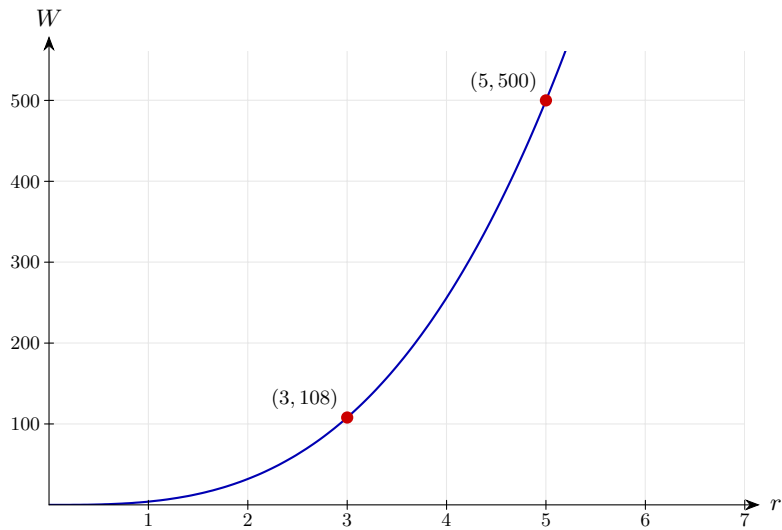
Step 2: Find W

$$W = 4 \times 125 = 500 \text{ g}$$

Cube $r = 5$ (giving 125) first, then multiply by k . Volume of a sphere scales with r^3 , so weight does too if density is fixed.

Final answer: 500 g

Graph.



The relationship plotted, with the data points marked.

Proportion 47 (cg4-047)

Difficulty: Foundation / Marks: 3

y is directly proportional to $(x + 1)^2$. When $x = 2$, $y = 27$. Find y when $x = 4$.

Worked solution.

Step 1: Find k

$$y = k(x + 1)^2 \implies 27 = 9k \implies k = 3$$

The whole bracket $(x + 1)$ is squared, so substitute $x = 2$ inside first: $(2 + 1)^2 = 9$. A common pitfall is to compute $x^2 + 1$ instead of $(x + 1)^2$.

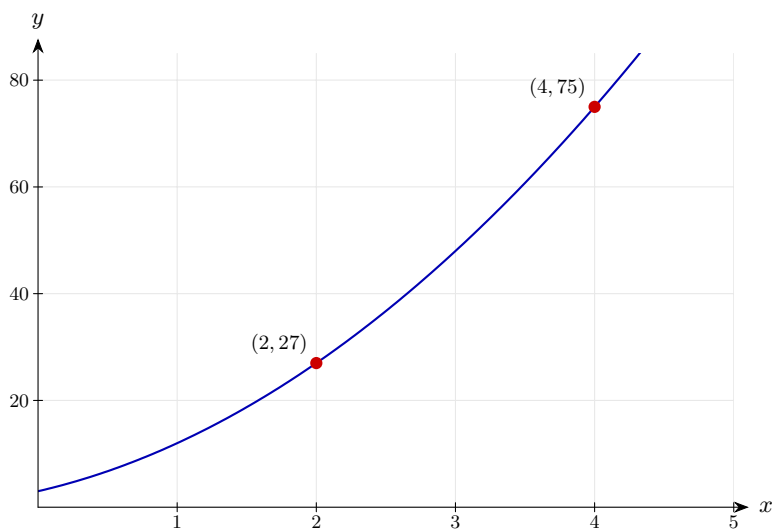
Step 2: Find y

$$y = 3 \times 25 = 75$$

With $x = 4$, the bracket is $(4 + 1)^2 = 25$. Add inside the bracket before squaring.

Final answer: $y = 75$

Graph.



The relationship plotted, with the data points marked.

Proportion 48 (cg4-048)

Difficulty: Foundation / Marks: 3

The intensity I of light is inversely proportional to the square of the distance d from the source. At 2 m, $I = 100$. Find I at 5 m.

Worked solution.

Step 1: Find k

$$I = \frac{k}{d^2} \implies 100 = \frac{k}{4} \implies k = 400$$

Inverse-square law: $I = \frac{k}{d^2}$. Square $d = 2$ to get 4, then multiply both sides by 4.

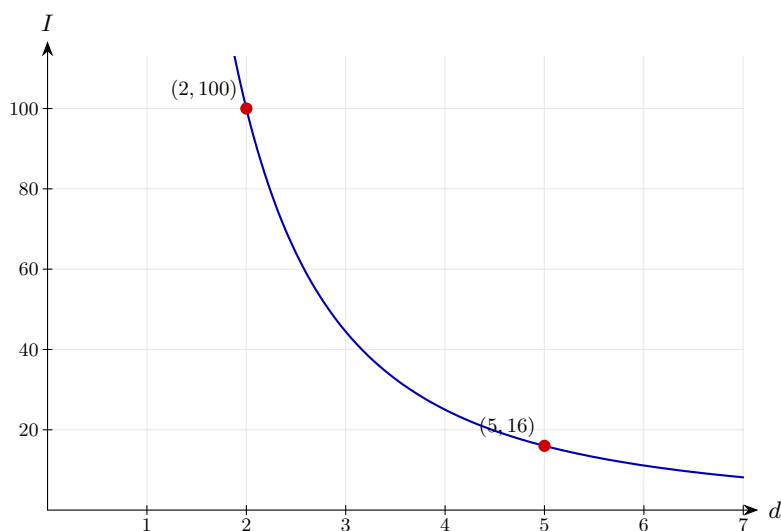
Step 2: Find I

$$I = \frac{400}{25} = 16$$

Square $d = 5$ (giving 25) before dividing. Distance went up by $2.5\times$, so d^2 went up by $6.25\times$ and intensity dropped by the same factor ($100 \rightarrow 16$).

Final answer: $I = 16$

Graph.



The relationship plotted, with the data points marked.

Proportion 49 (cg4-049)

Difficulty: Foundation / Marks: 4

Given $y = \frac{k}{x}$ and $y = 8$ when $x = 3$, find the value of x when $y = x$.

Worked solution.

Step 1: Find k

$$8 = \frac{k}{3} \implies k = 24$$

Substitute the known pair into $y = \frac{k}{x}$ and multiply both sides by 3 to isolate k .

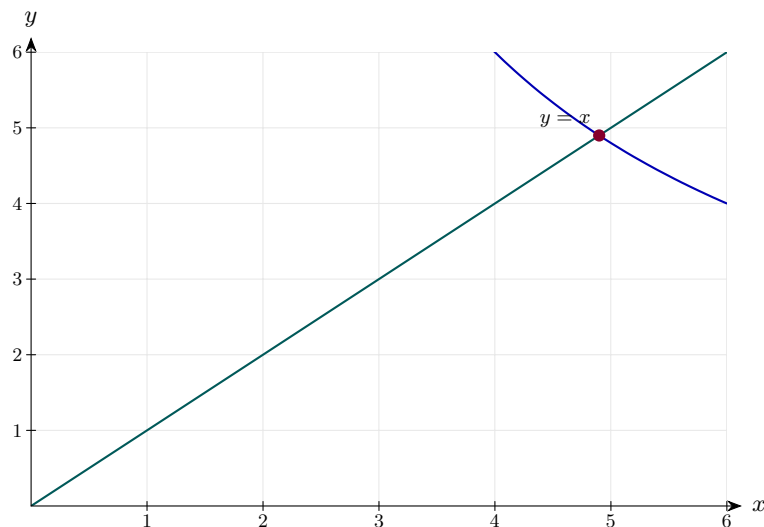
Step 2: Set $y = x$

$$x = \frac{24}{x} \implies x^2 = 24 \implies x = \sqrt{24} = 2\sqrt{6}$$

Replace y with x , then multiply both sides by x to clear the fraction. Simplify the surd: $\sqrt{24} = \sqrt{4 \times 6} = 2\sqrt{6}$; reject the negative root since the original equation $x = 24/x$ requires x and $24/x$ to share a sign.

Final answer: $x = 2\sqrt{6}$

Graph.



The relationship plotted, with the data points marked.

Proportion 50 (cg4-050)

Difficulty: Foundation / Marks: 3

y is proportional to x^n . When x is tripled, y is multiplied by 9. Find n .

Worked solution.

Step 1: Set up

$$y = kx^n; \quad k(3x)^n = 9kx^n \implies 3^n = 9$$

Write the original relation, then replace x with $3x$. The new y is 9 times the old, so divide both sides by kx^n — the constants cancel, leaving 3^n on its own.

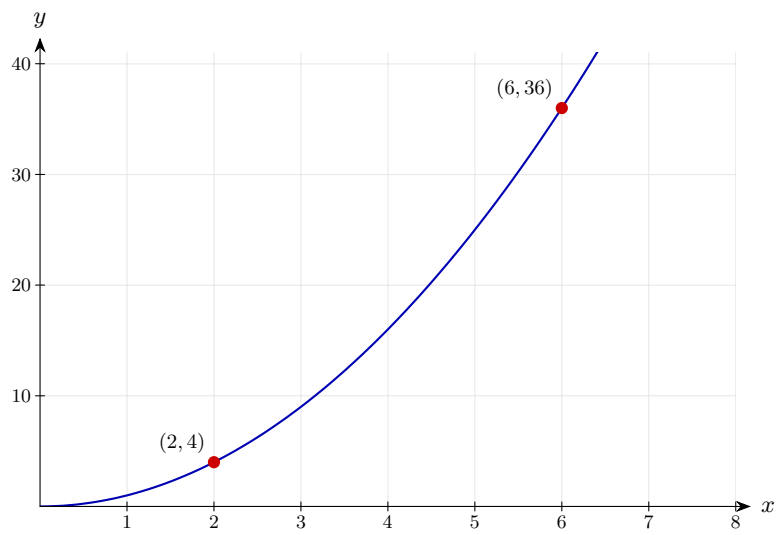
Step 2: Solve

$$3^n = 3^2 \implies n = 2$$

Write 9 as 3^2 . With matching bases, the exponents must be equal, so $n = 2$.

Final answer: $n = 2$

Graph.



The relationship plotted, with the data points marked.
